

If it is raining and John does not
have his umbrella with him
then he will get wet.

It is indeed raining.

John is not wet.

Therefore, John must be
having his umbrella with him.

If p and not q then r .
 p
 $\neg r$

} facts

Therefore, q .

|| deduce
↓
rigorous
↳ truth-preserving

Propositional Logic

Declarative sentences

→ as propositions

The number 5 is even.

The number 7 is prime.

Today is Tuesday.

Atomic
propositions
atoms

Reasoning calculus

↳ to draw conclusions
from a set of
assumptions.

→ in a reliable way
truth preserving

p :

The number 5 is even.

q :

The number 5 is prime.

negation
conjunction

$\neg p$
 $p \wedge q$

The number 5 is not even.

The number 5 is even
and the number 5 is prime.

disjunction
 $p \vee q$

The number 5 is even
or the number 5 is
prime.

implication
 $p \rightarrow q$

If the number 5 is even
then the number is prime.

$$(p \wedge q) \rightarrow (r) \vee \neg s$$

$$p \rightarrow q \rightarrow r$$

$$p \rightarrow (q \rightarrow r)$$

$$(p \wedge q) \rightarrow (\neg r) \vee s$$

$$p \wedge (q \vee r) \\ (p \wedge q) \vee r$$

Binding priorities

$\neg$ binds the most tightly.

$\wedge, \vee$ bind less tightly than $\neg$

but more than $\rightarrow$
 $\rightarrow$ is right associative

Natural Deduction

collection of proof rules that allows us to infer formulas from other formulas.

$$\begin{array}{ccc} \phi_1 & \phi_2 & \phi_3 \\ \hline (p \wedge \neg q) \rightarrow r, & \underline{p}, & \underline{\neg r} \end{array} \vdash q$$

a sequent is valid if a proof for it can be found. $\rightarrow$ derives

Rules for conjunction ($\wedge$)

$$\frac{\phi \quad \psi}{\phi \wedge \psi} \quad \wedge_i$$

$$\frac{\phi \wedge \psi}{\phi} \quad \wedge_{e_1}$$

$$\frac{\phi \wedge \psi}{\psi} \quad \wedge_{e_2}$$

$$p \wedge q, r \vdash q \wedge r$$

$$1. \quad p \wedge q$$

$$2. \quad r$$

$$3. \quad q$$

$$4. \quad q \wedge r$$

assumption / premise

premise

$\wedge e_2, 1$

$\wedge_i \quad 3, 2$

$(p \wedge q) \wedge r, s \wedge t \vdash q \wedge s$

1. $(p \wedge q) \wedge r$

2. $s \wedge t$

3. s

4. $(p \wedge q)$

5. q

6. $q \wedge s$

p.premise

p.premise

$\wedge e_1, 2$

$\wedge e_1, 1$

$\wedge e_2, 4$

$\wedge i \ 5, 3$

Rules for double negation

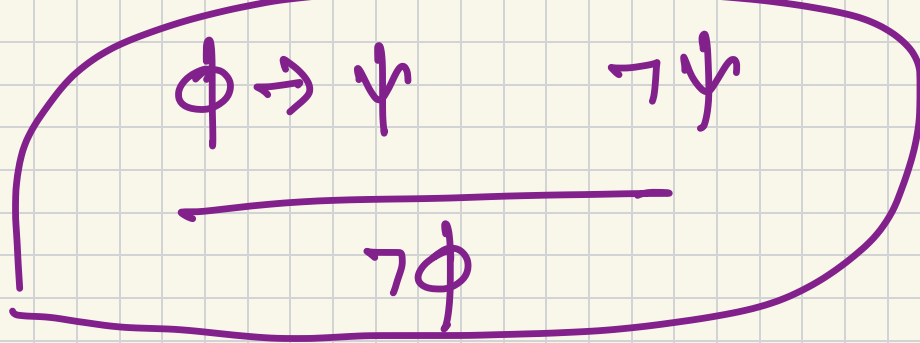
$$\frac{\phi}{\neg\neg\phi} \quad \neg\neg i$$

$$\frac{\neg\neg\phi}{\phi} \quad \neg\neg e$$

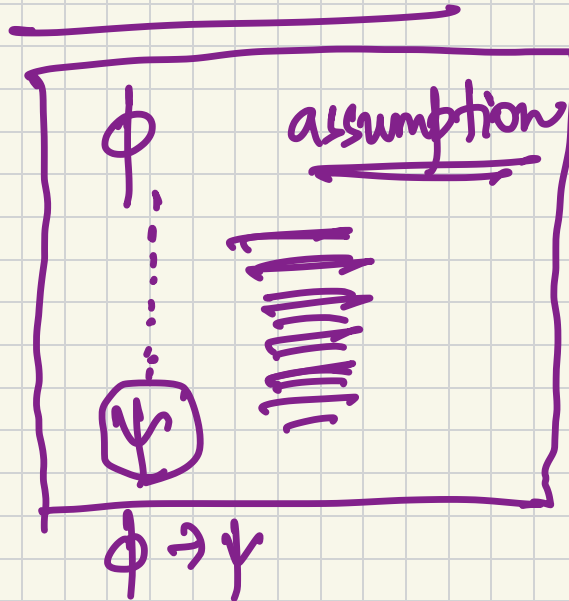
Rules for implication

(Modus ponens / MP)

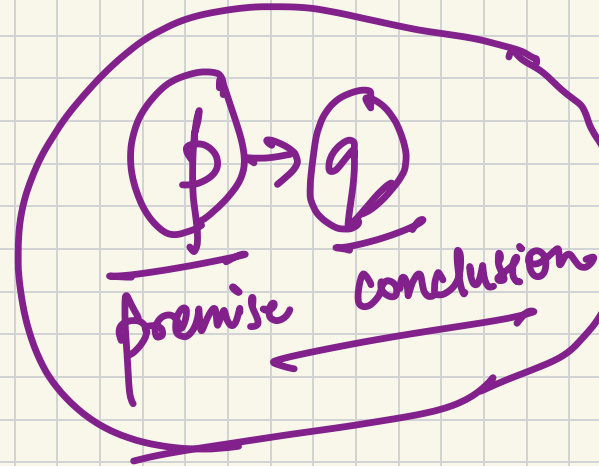
$$\frac{\phi \quad \phi \rightarrow \psi}{\psi} \quad \rightarrow e$$



Implication introduction



Modus tollens (MT)



$\rightarrow i$

$$\phi \rightarrow \psi \quad \vdash \quad \neg \psi \rightarrow \neg \phi$$

- | | | |
|----|-----------------------------------|---------------------|
| 1. | $\phi \rightarrow \psi$ | premise |
| 2. | $\neg \psi$ | assumption |
| 3. | $\neg \phi$ | MT 1, 2 |
| 4. | $\neg \psi \rightarrow \neg \phi$ | $\rightarrow i$ 2-3 |

$$\vdash (q \rightarrow r) \rightarrow ((\neg q \rightarrow \neg p) \rightarrow (p \rightarrow r))$$

1.	$q \rightarrow r$	assumption																																		
2.	$(\neg q \rightarrow \neg p)$	assum																																		
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